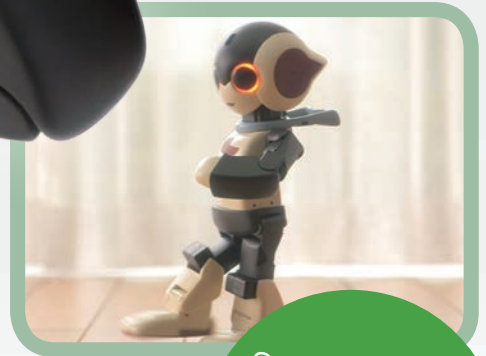
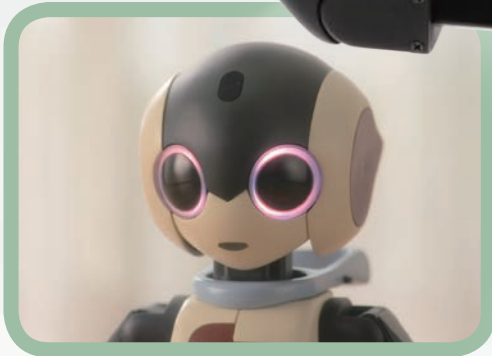
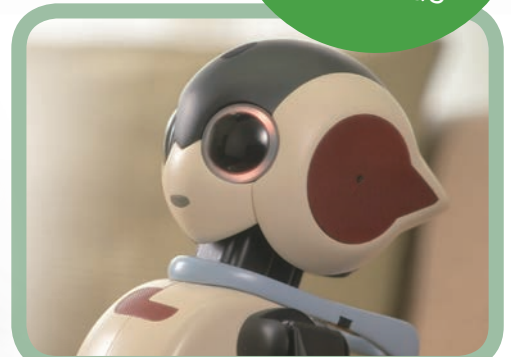


Build your own Robi

Pack **16**



Stages 59-62:
Begin working
on the inside of
Robi's head



Build your own **Robi**

CONTENTS

Assembly Guide **p320**

Stage 59: Test Robi's neck, back and arms	p320
Stage 60: Prepare and fit Robi's head servos	p328
Stage 61: Build up Robi's head	p338
Stage 62: Fit Robi's first infrared eye sensor	p342

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THIS PRODUCT IS NOT A TOY AND IS NOT DESIGNED FOR USE IN
PLAY. ITEMS MAY VARY FROM THOSE SHOWN. THERE MAY BE
COLOUR VARIATIONS OWING TO THE MATERIALS USED.

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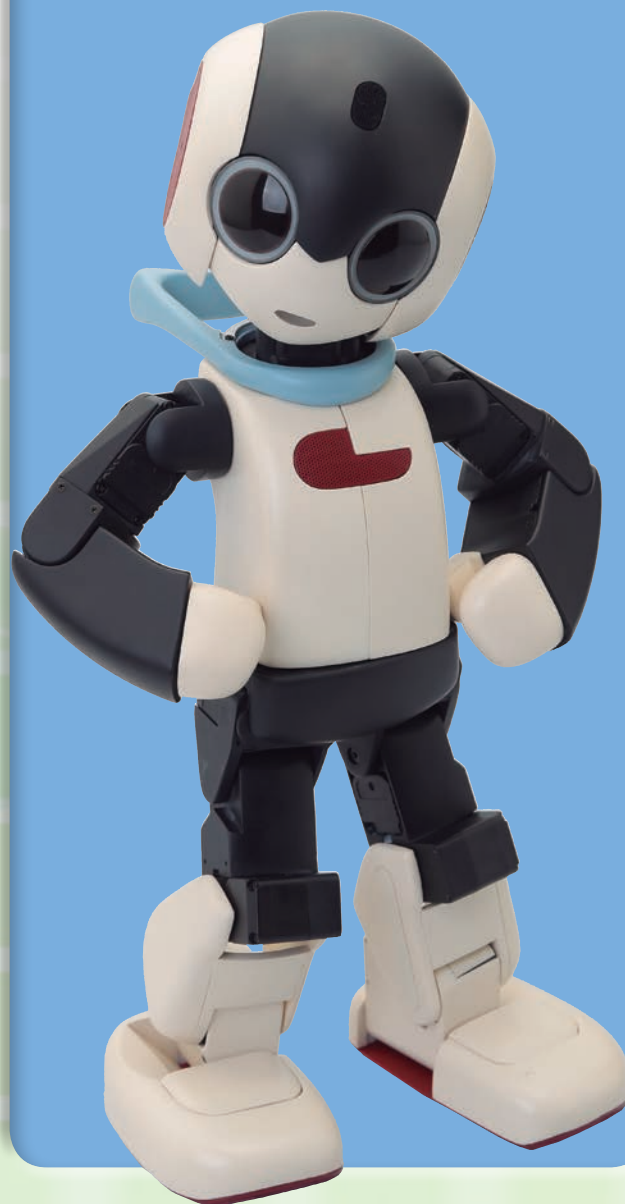
Published in the UK by De Agostini UK Ltd,
Battersea Studios 2, 82 Silverthorne Road,
London SW8 3HE

Published in the USA by De Agostini Publishing
USA, Inc., 121 E. Calhoun Street
Woodstock, IL 60098

Packaged by Continuo Creative,
39-41 North Road, London N7 9DP

Printed in EU

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STAGE 59: TEST ROBI'S NECK, BACK AND ARMS

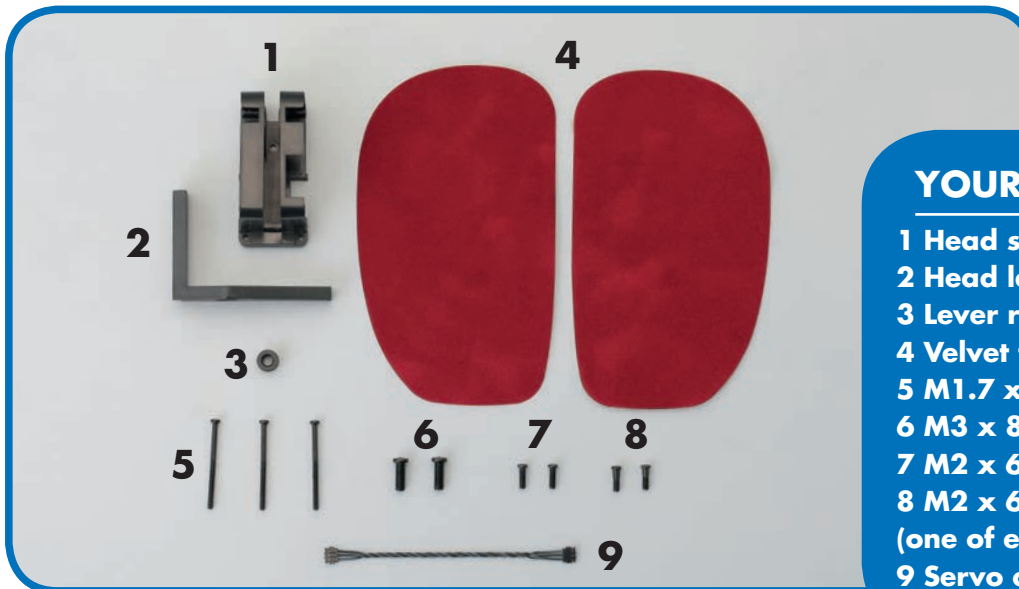
Use the CPU board to test Robi's neck, body and arms, before fitting the velvet pads supplied in this stage to his feet.

This stage begins with you testing all the servos of Robi's neck, back and arms together, to ensure they all work before you seal them inside Robi's body. You will then start building up the frame that will hold both of Robi's head servos, by securing this to the neck base and servo

assembly via the servo itself. Finally, you will attach the velvet pads supplied in this stage to the soles of Robi's feet. Unlike the double-sided adhesive pads used in Pack 9, these pads come in a left and right shape, so make sure you use the correct one!



**PARTS TO BE
ASSEMBLED**



YOUR PARTS

- 1 Head servo frame
- 2 Head lever
- 3 Lever ring nut
- 4 Velvet foot pads
- 5 M1.7 x 24mm pan-head screws x 3
- 6 M3 x 8mm pan-head screws x 2
- 7 M2 x 6mm pan-head screws x 2
- 8 M2 x 6.2mm pan-head screws x 2 (one of each screw is a spare)
- 9 Servo cable 70mm

SAVED PARTS

You will be using parts from previous stages, so be sure to have these to hand before beginning.



*Body
(Stage 58)*



*Neck stand
(Stage 53)*



*Legs and hip base
(Stage 52)*



*Protective pads
(Stage 3)*



*Battery
(Stage 38)*



*Power distribution board
(Stage 40)*

FASTEN THE NECK PLATE



- 1 Position the circled end of the neck plate fitted in the previous stage over the hole in the scarf beneath.



- 2 Use an M2 x 6.2mm pan-head screw to secure the part, but do not tighten it too much, as the plate needs to be able to move. You received a number of screws with this stage, so make sure you use the correct one – this one was part number 8 in the list.

TEST THE SERVOS



- 3 Hold the body as shown, then push the switch to the ON position.

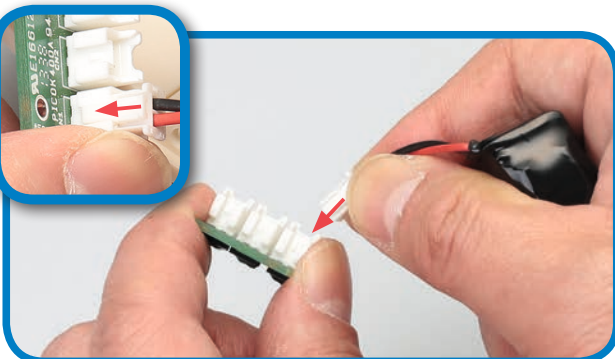
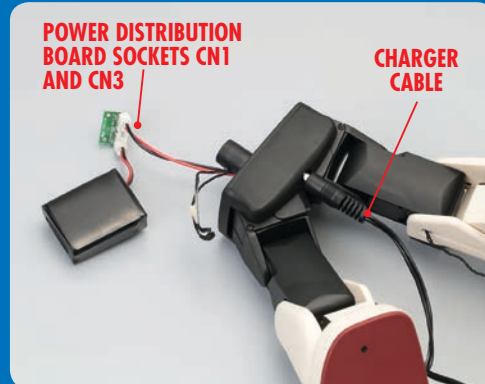
RECHARGING THE BATTERY



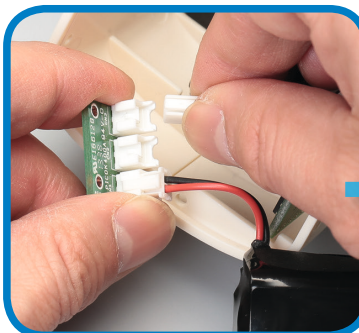
To recharge Robi's battery, connect the battery and power cord to the power distribution board, in sockets CN1 and CN3 respectively, then plug the mains charger into the back of the rear hip panel.

POWER DISTRIBUTION BOARD SOCKETS CN1 AND CN3

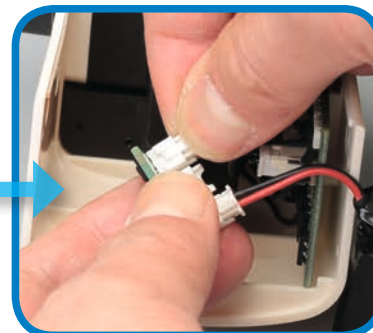
CHARGER CABLE

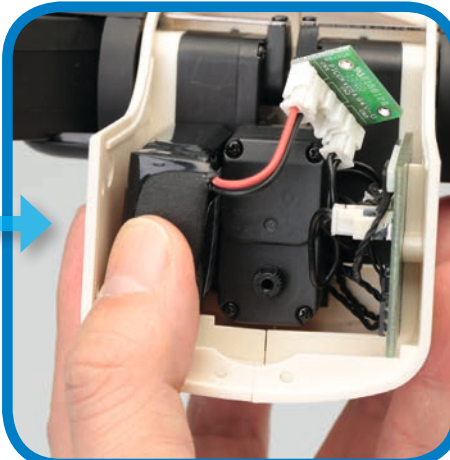


- 4 Connect the battery's connector to the socket marked CN1 on the power distribution board. Make sure the battery is charged (see box, above).



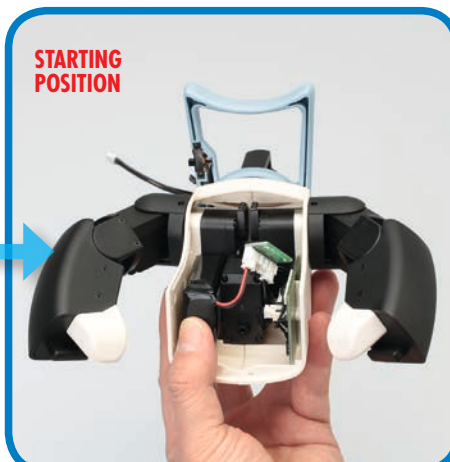
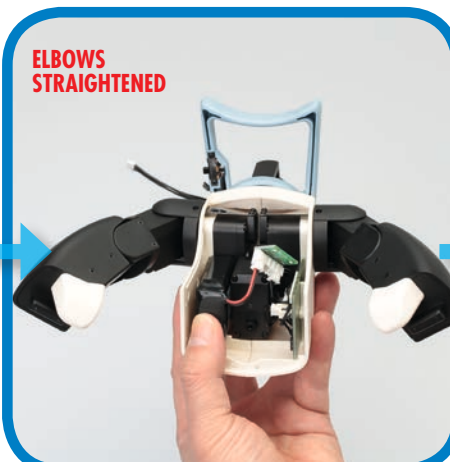
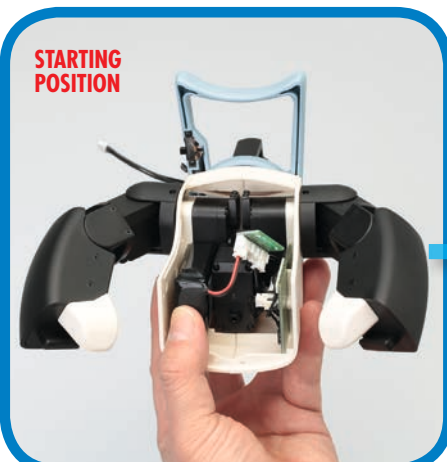
- 5 Now take the free connector on the body's power/switch cable and fit this into CN3.



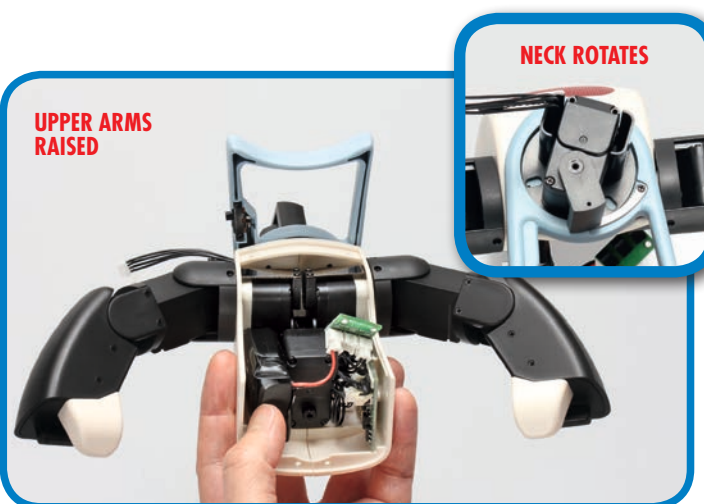


6 Slide the battery into the space next to the back servo, making sure all the connectors stay in their sockets. Hold the battery in place.

7 Still holding the battery in place, turn the switch ON. **Note:** Robi's arms will move as soon as the power is switched on!



8 When the power is connected, Robi will perform a movement test of his arms, by setting them to their starting, or neutral, position, stretching them outwards from the elbows, then bringing them back to the neutral position.

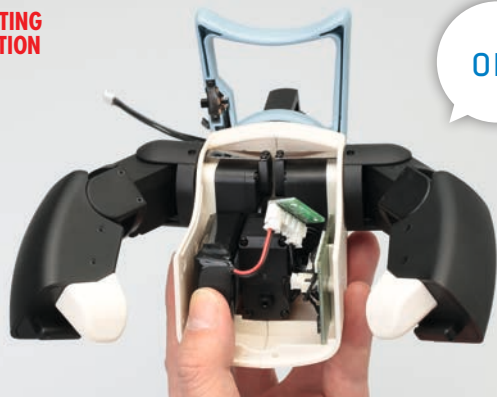


9 Next, Robi's upper arms will lift outwards – and while this happens, his neck servo will pan to the right and back.



10 Finally, Robi's shoulders will extend his arms forward over his chest – and the shaft of his back servo will rotate to the left.

STARTING POSITION

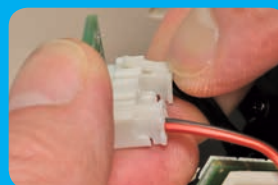


OK!

- 11** When the tests described on the previous pages are complete, Robi will return to his neutral position and you will hear him say 'OK!' This is to test his speaker. **Turn the switch back to OFF, and disconnect the switch cable and battery from the power distribution board.**

ATTENTION!

Each of the white plastic connectors has a small barb on one side, which holds the connector securely in its socket. To unplug a connector easily and safely, press the barb inwards – never pull on a connector without doing this!



IF THE ARMS DO NOT MOVE

Both arms were tested in Stages 43 and 44, so if they do not work now it is most likely a problem with the connection to the CPU, so check/reset this.

IF NO SERVOS MOVE

Remove the back servo's cable from the CPU and check to see if the arms start working.

IF THE BACK AND NECK DO NOT MOVE

Consult the troubleshooting guide of this stage.

Robi saying 'OK!' is only to test his speaker, not to affirm that the movement test went as planned. If the movement tests do not go as described, see the box above right, or the troubleshooting guide later in this stage.

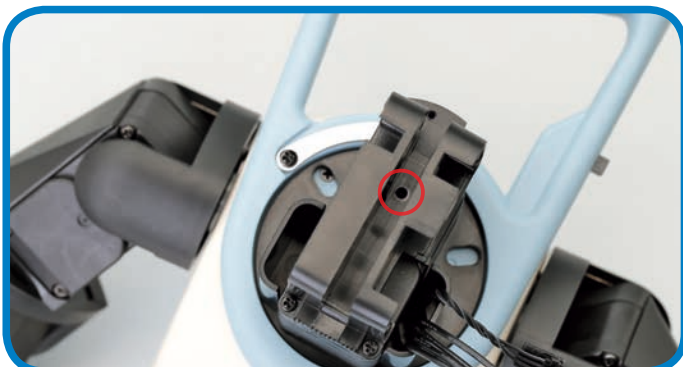
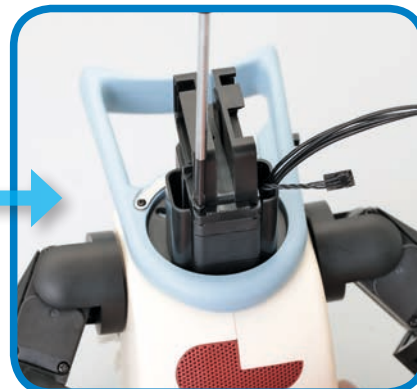
ATTACH THE HEAD SERVO FRAME



- 12** Line up the head servo frame to the neck servo, as shown.



- 13** Set the frame on top of the neck servo, so that the circled holes align with the free ones on the neck servo. Then insert and tighten two M1.7 x 24mm pan-head screws into these holes.

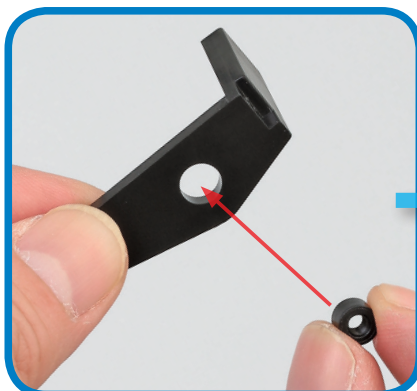


- 14** Now locate the circled hole in the top of the frame.



- 15** Insert an M2 x 6mm pan-head screw into this hole to secure the part.

FIT THE HEAD LEVER



- 16** Take the head lever and fit the ring nut into the hole in its longer arm, from the angle shown. The ring nut should fit perfectly.



- 17** Holding the ring nut in its hole, line up the lever to the head servo frame, oriented as shown above.



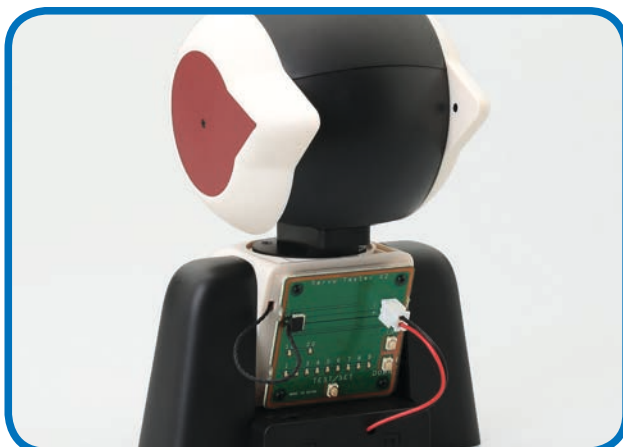
- 18** With the ring nut still in position, slide the lever into the space in the middle of the head servo frame. Its shorter arm should be at the rear of the assembly, pointing out towards Robi's right arm.



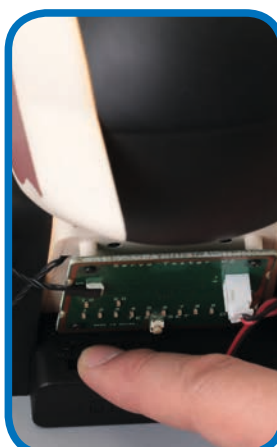
- 19** Take an M3 x 8mm pan-head screw and insert it through the circled hole in the inside of the head servo frame, entering from the left. Turn the screw through this hole so that it travels through the ring nut in the lever and out to the other side.



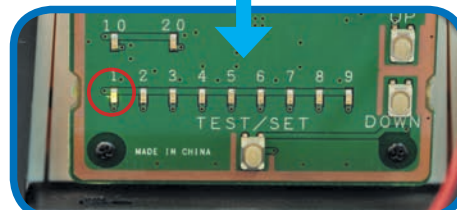
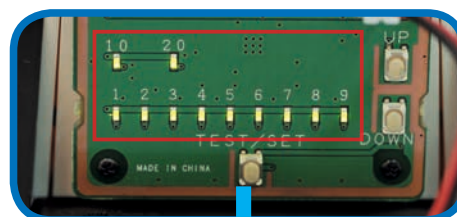
TEST AND FIT SERVO ID=15

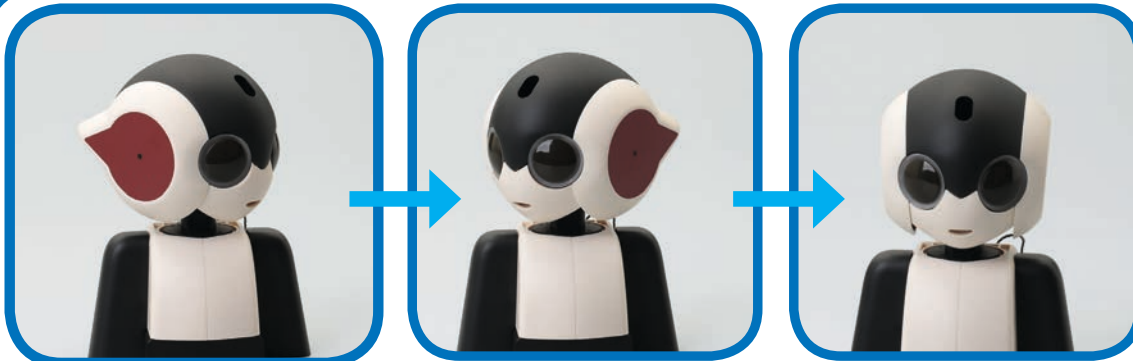


- 20** Make sure the test board is switched OFF, then connect its own servo cable to it.



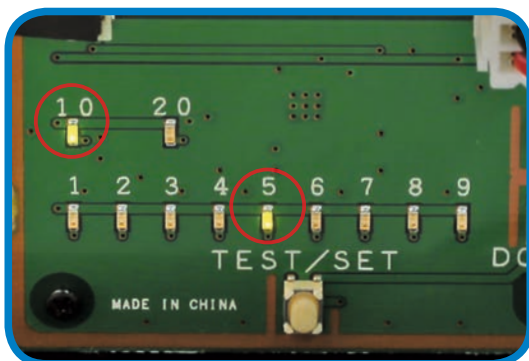
- 21** Turn the board ON. The LEDs should flash, and then leave the 1 LED lit alone.



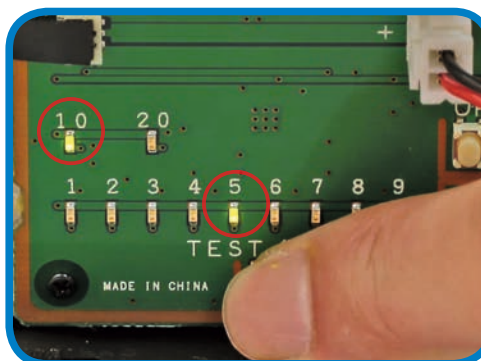


If the test does not run, check the servo cable's connector, and ensure that the battery is charged.

- 22** Press the TEST/SET switch: Robi's head should rotate 45° to the left, stop, then move back across to 45° to the right, before returning to its original position. The next step is to set the servo's ID number, so keep the power on. If the test does not run, or if the LEDs flash rapidly, refer to the HELP box in Stage 8.



- 23** Although this servo has been used for Robi's neck during the testing process, it will be his forward-tilting head servo when fitted, with the ID number 15. To set this, press UP 14 times, so that the 10 and 5 LEDs light up.



- 24** Press and hold down the TEST/SET button for a few seconds. The 10 and 5 LEDs should flash, then remain lit. The ID number is now set.

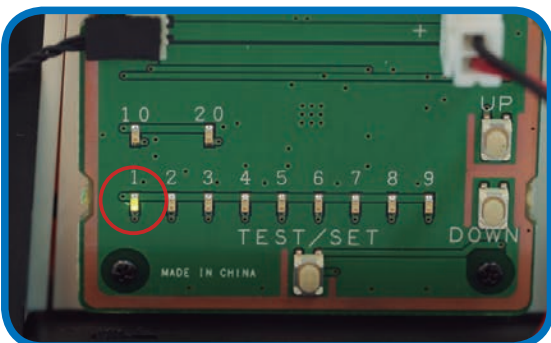


ID=15

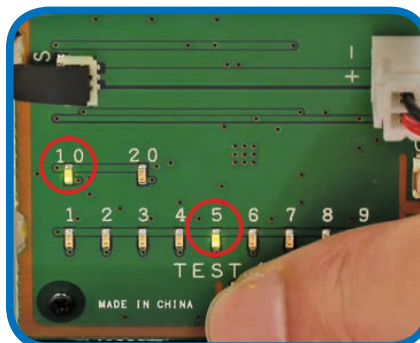


As you will have seen in the previous few packs, the process of checking and resetting a servo's ID number once fitted can be fiddly and time-consuming, so be sure to get it correct now!

CHECKING THE ID NUMBER



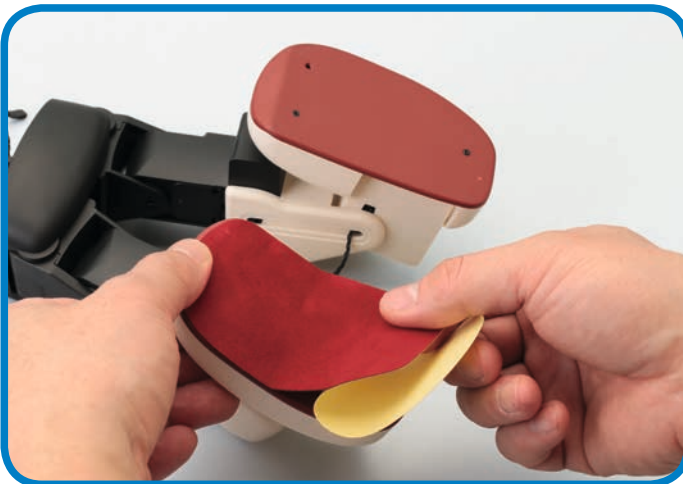
- 25** It is vital that Robi's servos are set correctly, so it is always a good idea to check that you have set the intended ID number afterwards. To do this, leave the servo connected, then switch the power OFF, then back ON. Only the 1 LED should light up.



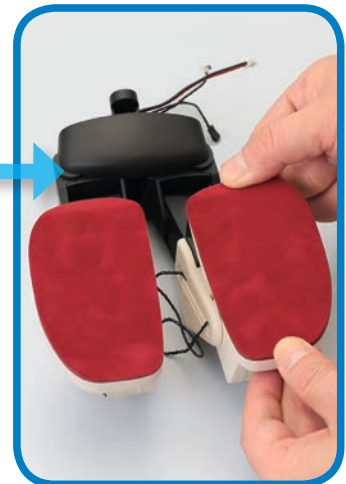
It is normal for the 1 LED to light up when you switch on. It will only change to give you the ID number of the servo when you press TEST/SET. If you made a mistake, or set the wrong ID number, you can reset it by going through the setting procedure again.

- 26** To check the ID number, press TEST/SET briefly. Now the 10 and 5 LEDs should light up and the servo shaft should repeat the rotations seen in the test. Note: if anything is amiss, you can begin the setting process again from Step 23.

FITTING THE FOOT PADS

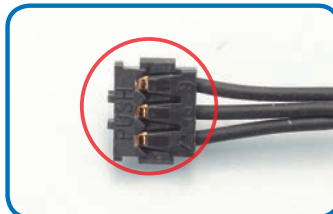
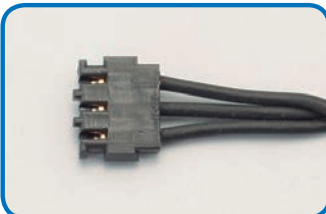


- 27** Match the right foot pad to the sole of Robi's right foot, then, when you are sure that they align, peel the backing away to the middle. Press the heel of the pad onto the back of the sole, then smooth it down from the back to the front, peeling the rest of the backing away as you do.



- 28** Make sure that the pad fits exactly: if it doesn't, peel it away and reset it. Once you are happy with it, repeat the process for the left foot and pad.

PREPARING THE SERVO CABLE



- 29** As you have done previously, identify the side of this stage's servo cable connector that has PUSH marked on it (circled).



- 30** Apply a pad to the connector on this side. Repeat for the other end of the cable.

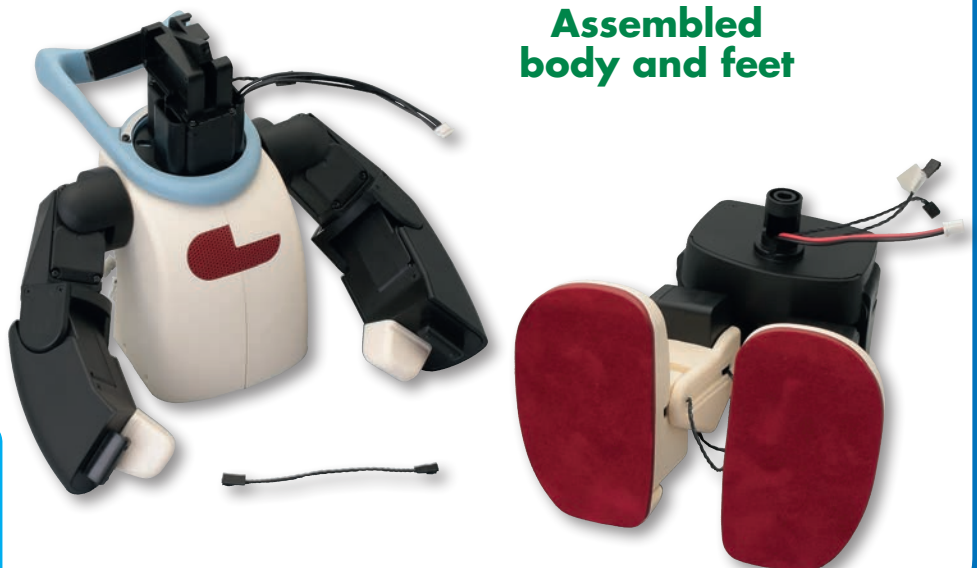
If you want to delay fitting the pads until the cable is used in the build, you may do so.



Remember, to see all of Robi's assembly stages, including this one, visit the ModelSpace web page at:

www.model-space.com

Assembled body and feet



TROUBLESHOOTING

In this section, we look again at what to do if the movement test carried out in this stage does not go as described. As the arms have been tested previously, we will focus on the operation of the neck and back servos, as well as the speaker, from which Robi says 'OK!'

Along with the Phillips screwdriver supplied with Stage 2, you will need the following:

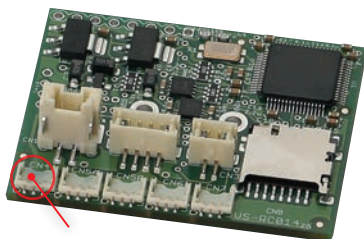


Neck stand (Stage 53)

INITIAL CHECKS

Before dismantling the assembly to check its individual servos, check the other electrical components used in the test.

CHECK THAT THE CABLE IS CONNECTED TO THE CORRECT SOCKET ON THE CPU



HEAD/NECK/BACK SOCKET

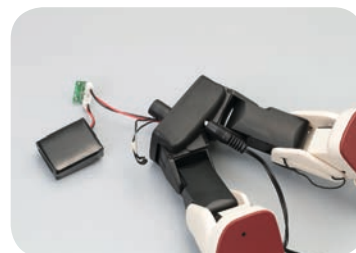
The socket farthest from the micro SD card slot on the CPU board will hold the servo cable running from the back servo.

CHECK THAT THE SERVO CABLE IS SECURELY CONNECTED TO THE CPU



Ensure that the back servo cable's connector is sitting securely within its socket. Remove and replace it carefully if need be.

CHECK THAT THE BATTERY IS CHARGED



The test will only run if sufficient power is received from the battery, so make sure the battery is fully charged (see the Close-Up box earlier in this stage).

IF NEITHER THE NECK NOR BACK SERVOS MOVE, OR IF ROBI DOES NOT SAY 'OK!'

If neither the neck nor the back servo moves when you start the test, it is probable that there is a misplaced connection, so go back and check these. Similarly, if Robi does not say 'OK!' at the end of the test, you should check the connection between the speaker cable and CPU board.



THE BACK SERVO

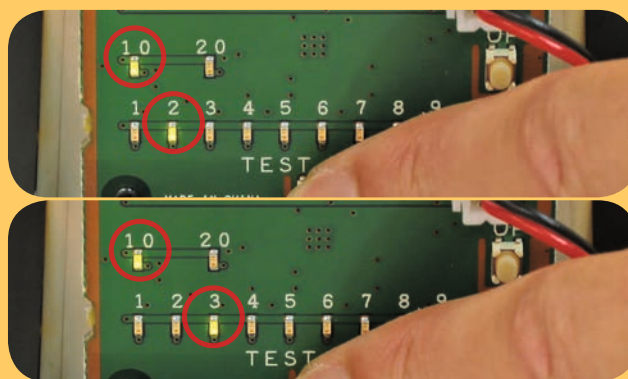
Follow the assembly of Steps 14-19 of Stage 58 in reverse, and check the back servo's connections carefully. If both were sound, disconnect it from the neck servo and check the servo's ID number (ID=12), resetting it if necessary.

THE SPEAKER

If Robi does not say 'OK!' at the end of the test, check the connection of the speaker cable to the CPU – done in Step 15 of Stage 57.

IF ONLY ONE SERVO MOVES

If one servo moves but the other does not, it is likely that one of the ID numbers has been set incorrectly.



Should you need to disassemble and reset Robi's neck or back servos, see Stage 58 (for the back servo, ID=12) and/or Stage 54 (for the neck servo, ID=13).

STAGE 60: PREPARE AND FIT ROBI'S HEAD SERVOS

Prepare Robi's second head servo then fit it, along with the one from the previous stage, to his head servo frame.

There are two servos inside the upper part of Robi's head, ID=14 and ID=15. Both are set onto the head servo frame that you fitted and wired over the previous stages, and they will work in tandem with the internal lever and

head base to make Robi's head tilt forward and back (in the case of ID=15) and side to side (ID=14). As ever, take care when handling open servos, and make sure you set the correct ID number.

PARTS TO BE ASSEMBLED



1



YOUR PARTS

1 Servo motor (ID= 14)

SAVED PARTS

You will be using parts from the previous stages, so be sure to have these to hand before beginning.



Neck stand
(Stage 53)



Battery
(Stage 38)

Power distribution board
(Stage 40)



Body and neck assembly, and
servo cable (Stage 59)

THE SERVO AND CABLE

TIP!

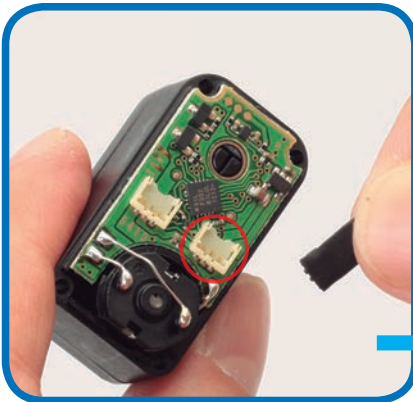


As ever, it is very important that the electronic parts exposed by removing the servo's cover are not touched in any way. Make sure to hold the open servo by its sides, and if you put it down on your work surface during the assembly, do this with the open side facing upwards.

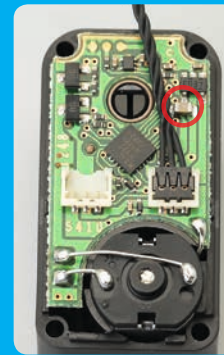


1 Take out the four screws holding the servo's removable cover in place, and keep these safely to one side.

2 Gently lift the cover away, taking care not to force it or damage the circuit board underneath.

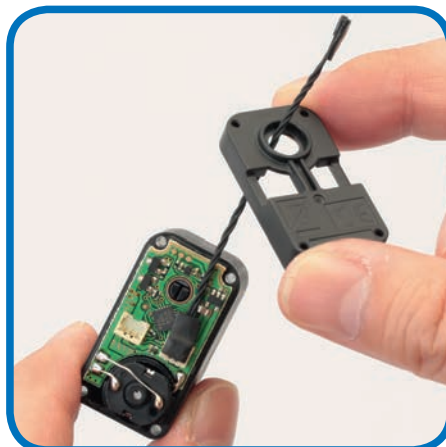


TIP!



This servo cable had protective pads fitted to its connectors in the previous stage, but if you find it difficult to attach with a pad in place, it is possible to fit the pad afterwards. If you choose to do the assembly this way, be especially careful not to press down on the circled projection, next to the cable, as you do so.

3 Hold the open servo with the circuit board facing upwards. Fetch the servo cable fitted with protective pads in the previous stage, and press one of its connectors into the socket circled in red, as you have done before. It does not matter which end of the servo cable you use.

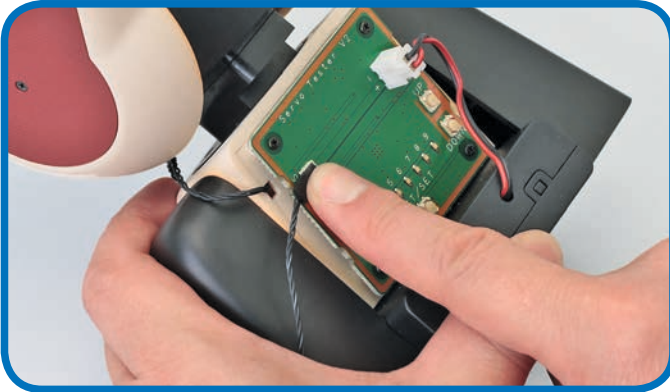


4 Pass the free end of the servo cable through the circular hole in the removable cover, as shown.

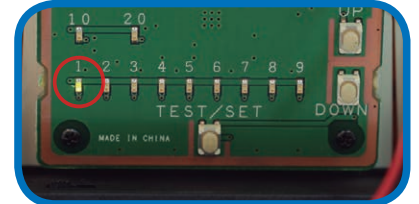
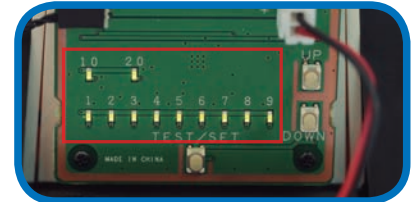
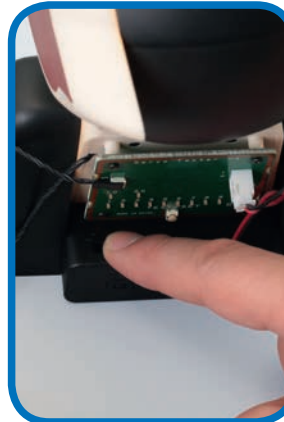
5 Press the cover back into place on the servo's main housing.

6 Fasten the cover back into place with all four screws, but only lightly, as you will be removing them again at a later stage.

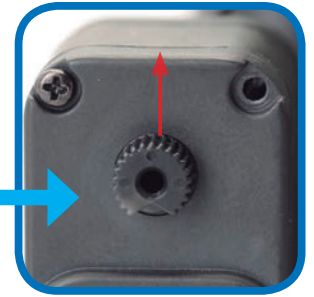
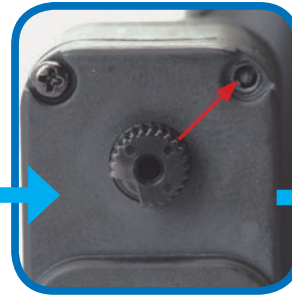
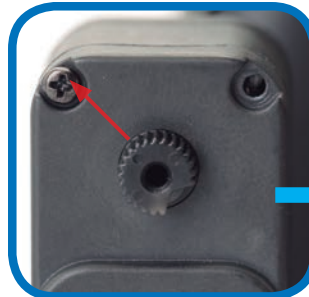
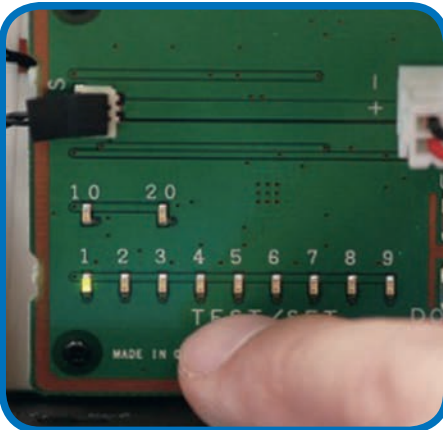
TESTING THE SERVO



- 7** Take the neck stand and make sure the power pack is turned OFF. Carefully unplug the black servo cable from the socket on the left of the test board, and attach the servo cable connected to the servo in Step 3.

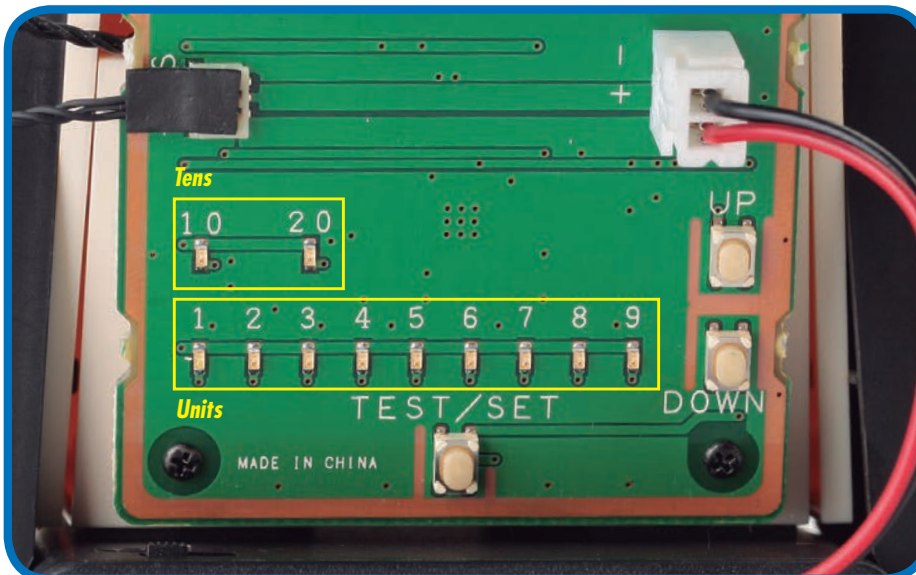


- 8** With the servo connected, switch the power to ON. After the LEDs blink twice, only the 1 LED should light up.

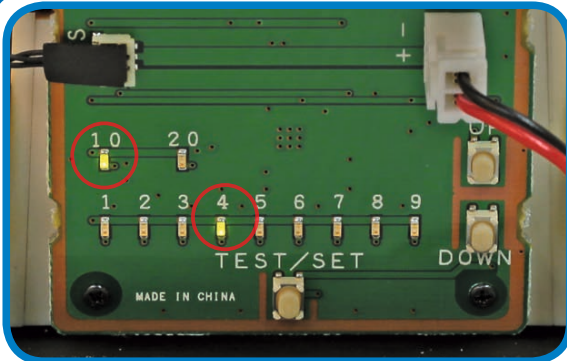


- 9** To test the servo, press the TEST/SET switch: the servo's shaft should rotate 45° to the left, stop, then move back across to 45° to the right. The next step is to set the servo's ID number, so keep the power on. If the test does not run, or if the LEDs flash rapidly, refer to the HELP box in Stage 8.

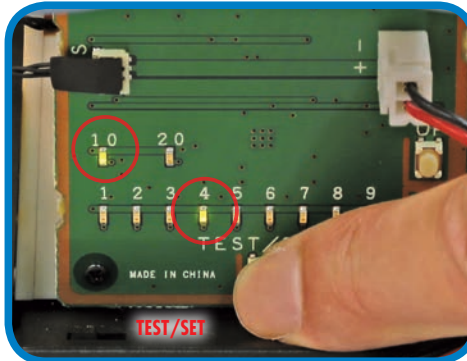
SETTING THE ID NUMBER



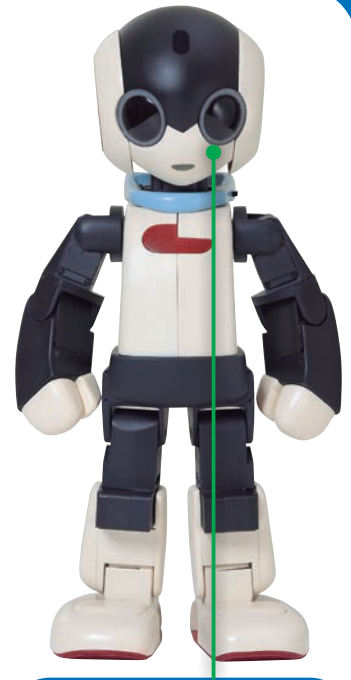
- 10** Next, you will set the ID number of the servo using the test board, as you have done before. Each time you press the UP switch, the ID number increases by one, and the result is displayed by the two banks of LED lights. The top row reads in tens and the bottom in units. For example, to set the ID to 14 as you will do in this stage, you press UP 13 times. This will light up the 10 and 4 LEDs. (If you overshoot, press the DOWN switch to take it down one step at a time.) When the ID number is correct, you set it by pressing and holding the TEST/SET switch.



11 To set the head servo to ID=14, press UP 13 times, so that the 10 and 4 LEDs are lit. Pressing DOWN will adjust this if you overshoot while doing this.

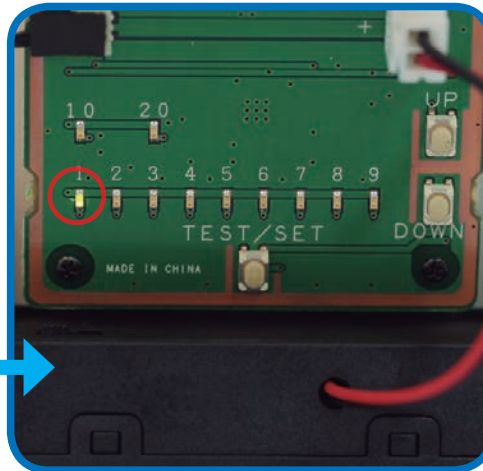
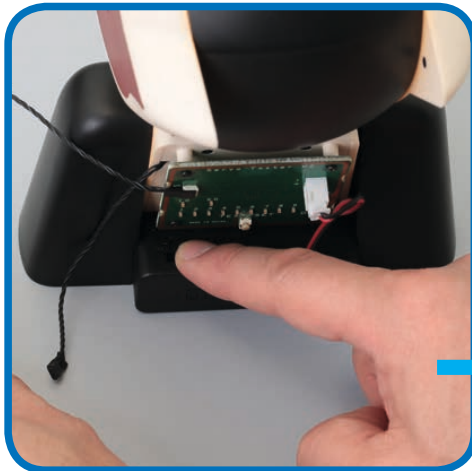


12 Once the numbers are correct, press and hold TEST/SET for a few seconds. The LEDs will blink, then stay lit, confirming that the ID is set.

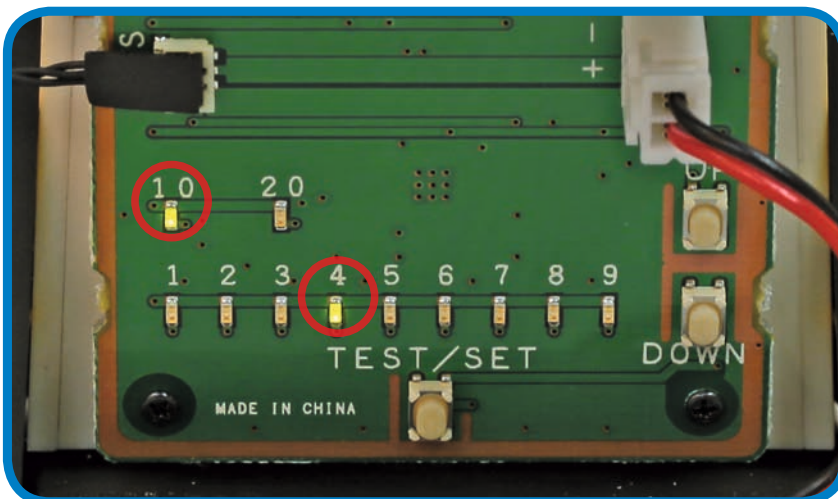


ID= 14

It is normal for the 1 LED to light up when you switch on. It will only change to give you the ID number of the servo when you press TEST/SET. If you made a mistake, or set the wrong ID number, you can reset it by going through the setting procedure again.

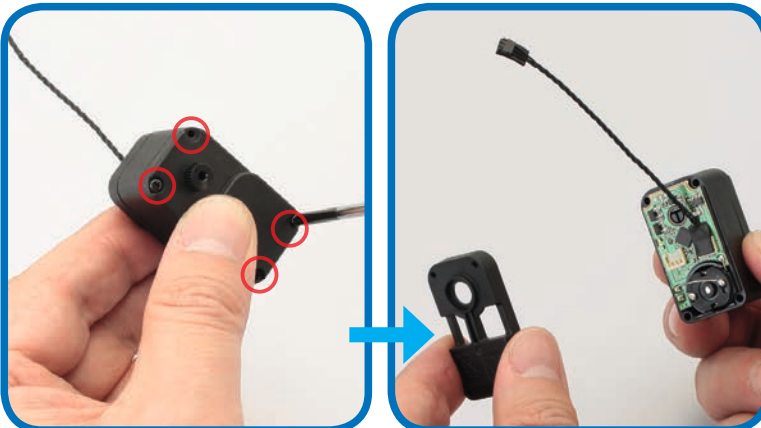


13 It is vital that Robi's servos are set correctly, so it is always a good idea to check that you have set the intended ID number afterwards. To do this, leave the servo connected, then switch the power OFF and back ON again. After blinking, only the 1 LED should light up.



14 To check the ID number, press TEST/SET briefly. Now the 10 and 4 LEDs should light up and the servo shaft should repeat the rotations seen in the test. Note: if you find that a servo has the wrong number assigned to it, go back and repeat the setting process in full.

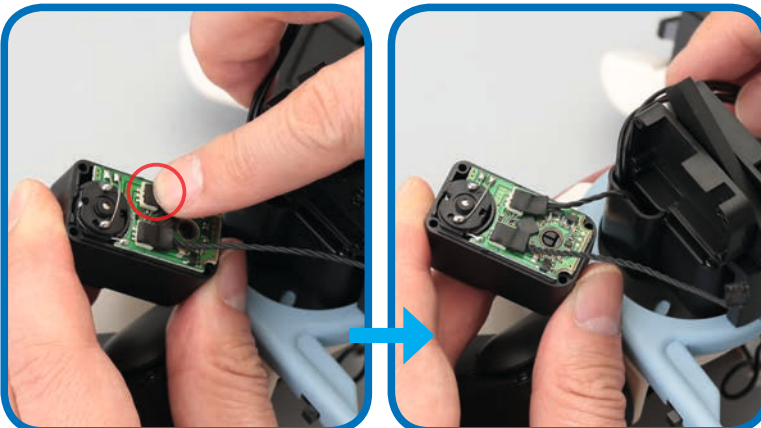
ATTACH THE FIRST HEAD SERVO



15 Remove all four screws again from the servo prepared in this stage (ID=14) and remove its cover. You will not be using the cover again.



16 Line up the servo to the left-hand side of Robi's neck base. Take the servo cable leading from the neck base in your hand.



17 Set the neck servo's cable connector into the free socket on the head servo ID=14.



18 Run the four-strand VR cable, also leading from the neck base, into the larger, circled cutaway in the left side of the head servo frame.



19 With the VR cable still in the larger cutaway, bring the open face of the servo towards the head frame, with the servo shaft at the rear of the assembly, closest to Robi's scarf.



20 Make sure the VR cable is still in the larger cutaway, then run the servo cable into the smaller cutaway in the frame. Be careful not to disturb the connectors on the servo's face as you do this, or to trap or pinch any of the cables themselves.



- 21** Join the servo to the frame – checking again as you do so that none of the cables are trapped, and that the connectors stay in place.

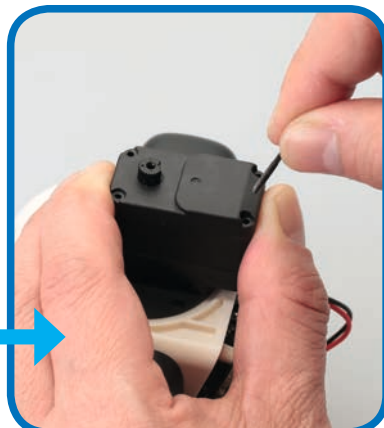


- 22** Use all four screws removed in Step 15 to secure the servo to the head frame.

REMOVING SERVO ID=15



- 23** Carefully remove Robi's head from the neck stand.



- 24** Remove all four screws holding the neck servo in place. Keep these, as you will be using them again.

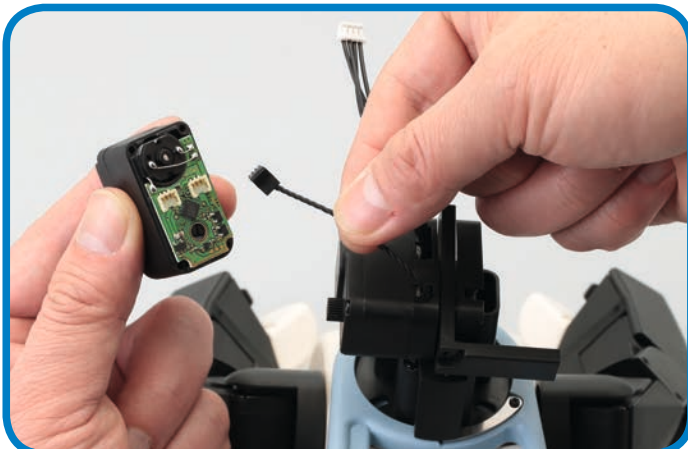


- 25** Gently lift the servo away from its base, but do not move it far enough to disconnect the servo cable.



- 26** Now unhook the servo cable by hand to free servo ID=15.

ATTACH THE SECOND HEAD SERVO



- 27** Position servo ID=15 next to the free side of the head frame, on the right side of Robi's neck base. Line up the servo cable leading to servo ID=14, fitted in this stage, to its open face.



- 28** Connect this servo cable to one of the sockets. The other will not be used.



- 29** Bring the servo to the frame, and pass the servo cable connected in the previous step through the circled cutaway.



- 30** Join the servo to the frame, making sure the servo cable is not trapped and remains neatly in the cutaway. Pulling gently upwards as you combine the parts will achieve this.



- 31** Use all four screws removed in Step 24 to secure the servo to the head frame.



ROBI'S HEAD, NECK STAND AND TEST BOARD



Although you have disassembled the neck stand in recent stages, you will be using it and the test board that it holds again, so keep it safely until then!



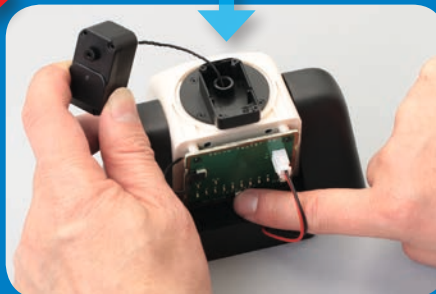
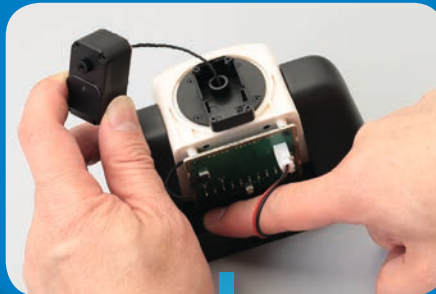
Robi's head

Robi's head contains a number of his most vital electronic components, which you will begin to add in the coming stages. The first components you will fit, in Stage 62, are the infrared sensors that identify movement and are set into the centre of his eyes. Then, later on, you will add the LEDs that illuminate his eyes and light up his mouth to give him his expressive character. The head will also hold key parts used for his voice-recognition capabilities, and more!



USING THE NECK STAND AND TEST BOARD

Although you have removed the scarf, neck servo and head, you will continue using the neck stand, and the test board it holds, to check or reset servos' ID numbers. The process for doing this will be the same as before, though is actually a little more straightforward, as you can now use the neck stand's existing servo cable (previously used to connect the neck servo) to connect to the servo in question. The steps for performing a servo test this way are found below.



1 Ensure the neck stand's servo cable is connected securely to the test board, then set the other end to the servo you wish to check/reset.

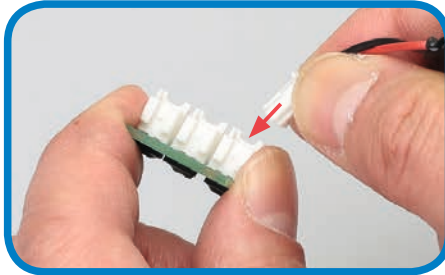
2 Begin the process of testing/resetting the chosen servo in the usual way.

3 To complete the test, simply turn the test board OFF and carefully disconnect the servo.

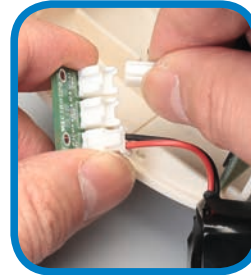
TEST THE NECK AND HEAD SERVOs



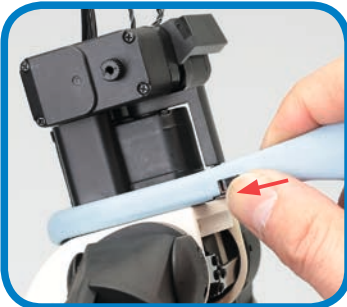
32 Push the switch on the scarf to the OFF position.



33 Connect the battery to the power distribution board, in socket CN1.



34 Now connect the power/switch cable to socket CN3.



35 Turn Robi's switch ON. The test will begin as soon as power is turned on, so be prepared!
As well as the servos focused on in these steps, the arms will also move, so hold Robi in a way that will not interfere with this.

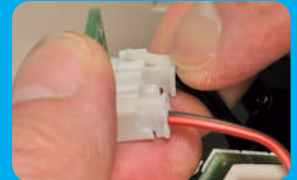


36 The test will begin once the power is turned on. First, observe the left-hand head servo, ID=14, which will return to its neutral position, then rotate a small way clockwise, raising the D-cut section upwards. Look very closely, as the movement will be minimal (inset). Robi's neck servo, ID=13, will then rotate. Once that is done, Robi will say 'OK!'



ATTENTION!

Each of the white plastic connectors has a small barb on one side, which holds the connector securely in its socket. To unplug a connector easily and safely, press the barb inwards – never pull on a connector without doing this!



37 While the neck servo is performing its rotation, Robi's second head servo, ID=15, will also run its test, this time rotating *anticlockwise* a little to again raise its D-cut upwards (inset). Once you are happy with the test, disconnect the power.



Robi saying 'OK!' is only to test his speaker, not to affirm that the movement test went as planned. If the movement tests do not go as described, see the box in Stage 59, or the troubleshooting guide later in this stage.

Assembled body and head servos



Remember, to see all of Robi's assembly stages, including this one, visit the ModelSpace web page at:

www.model-space.com

TROUBLESHOOTING

In this section, we look again at what to do if the movement test carried out in this stage does not go as described. As the arms have been tested previously, we will focus on the operation of the neck and back servos, as well as the speaker, from which Robi says 'OK!'

Along with the Phillips screwdriver supplied with Stage 2, you will need the following:

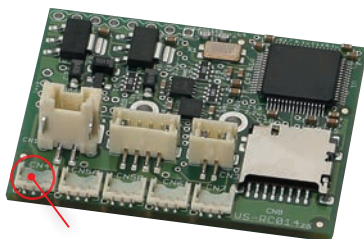


Neck stand (Stage 53)

INITIAL CHECKS

Before dismantling the assembly to check its individual servos, check the other electrical components used in the test.

CHECK THAT THE CABLE IS CONNECTED TO THE CORRECT SOCKET ON THE CPU



HEAD/NECK/BACK SOCKET

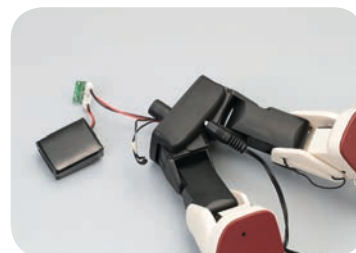
The socket farthest from the micro SD card slot on the CPU board will hold the servo cable running from the back servo.

CHECK THAT THE SERVO CABLE IS SECURELY CONNECTED TO THE CPU



Ensure that the back servo cable's connector is sitting securely within its socket. Remove and replace it carefully if need be.

CHECK THAT THE BATTERY IS CHARGED



The test will only run if sufficient power is received from the battery, so make sure the battery is fully charged (see the Close-Up box in Stage 59).

IF NEITHER HEAD SERVOS MOVES

If neither head servo moves when you start the test, it is probable that there is a misplaced connection, so go back and check these. If this does not resolve the issue, you will need to check/reset their ID numbers.

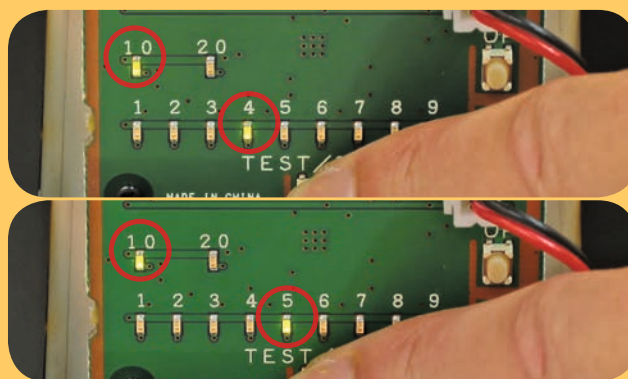


DISASSEMBLY

Follow the assembly of Steps 15-31 of this stage in reverse, and carefully check the connections made in the steps before. If both were sound, you will need to check the servos' ID numbers, resetting them if necessary.

IF ONLY ONE SERVO MOVES

If one servo moves but the other does not, it is likely that one of the ID numbers has been set incorrectly.



If you need to reset Robi's head servos, they are ID=14 (left-hand side of head servo frame, programmed in this stage) and ID=15 (right-hand side of frame, programmed in Stage 59).

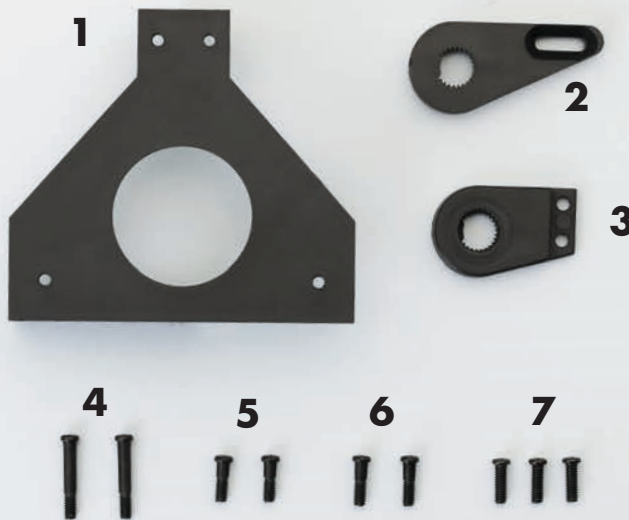
STAGE 61: BUILD UP ROBI'S HEAD

Add two servo horns to the head servos fitted over the previous stages, then combine these with the base that will hold the electronic components enclosed in Robi's head.

The two servo horns supplied in this stage link up with the head base, using the rotation of the head servos' shafts to angle it forward and back (via the head lever, in the case of the right-hand servo, ID=15) and side to side (using the long screw and elongated

hole in the horn, in the case of servo ID=14). Two of the screws are 'stepped', meaning they have a section of unthreaded shaft that enables the joints they are used for to pivot freely – because of this, they must not be overtightened.

PARTS TO BE ASSEMBLED

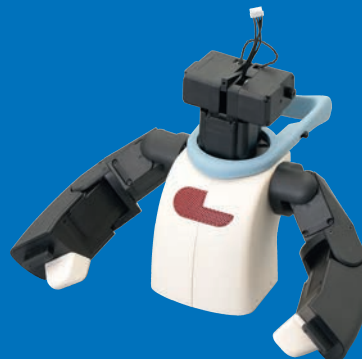


YOUR PARTS

- 1 Head base
- 2 Servo horn (sideways tilt)
- 3 Servo horn (forward tilt)
- 4 M2 x 12mm pan-head screws x 2
- 5 M2 x 6.2mm pan-head screws x 2
- 6 M2 x 6.7mm pan-head screws x 2
- 7 M2 x 6mm pan-head screws x 3 (one of each screw is a spare)

SAVED PARTS

You will be using parts from the previous stages, so be sure to have these to hand before beginning.



Body, neck and head servo assembly (Stage 60)

FITTING THE SERVO HORNS



- 1** There are two servo horns supplied in this stage: part 2, with the elongated body, will tilt Robi's head from side to side, and will fit onto the servo shaft of servo ID=14, on the left side of Robi's head. **The parts will join via their D-cuts (inset), with the horn pointing backwards, so, as ever, make very sure that you fit these correctly.**



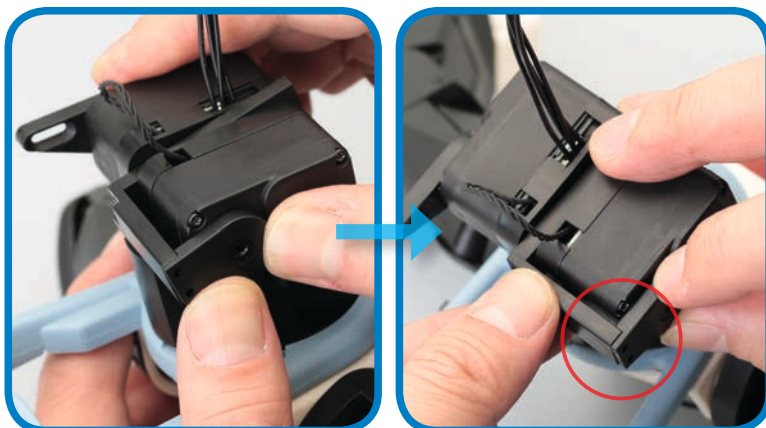
- 2** Fix the servo horn to the shaft, making sure the D-cuts fit together easily and correctly.



- 3** Now line up the other, shorter-bodied servo horn (part 3) to the right hand side of Robi's head. Its D-cut will fit it to the shaft of the right-hand head servo, ID=15.



- 4** Fix the part to the servo shaft. The ridge at the square edge of the part should combine with that of the lever fitted previously, with the circled holes aligning.

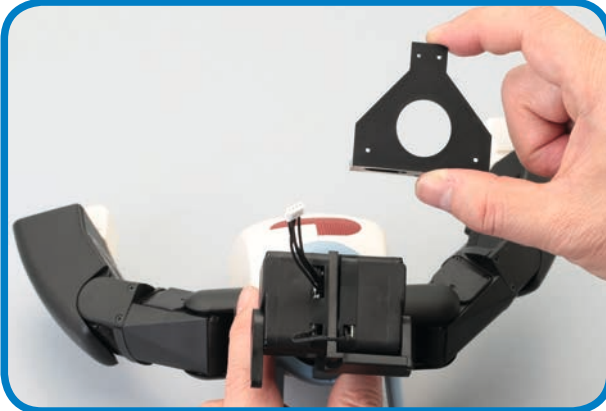


- 5** Press the parts together, ensuring they fit together neatly, as shown.

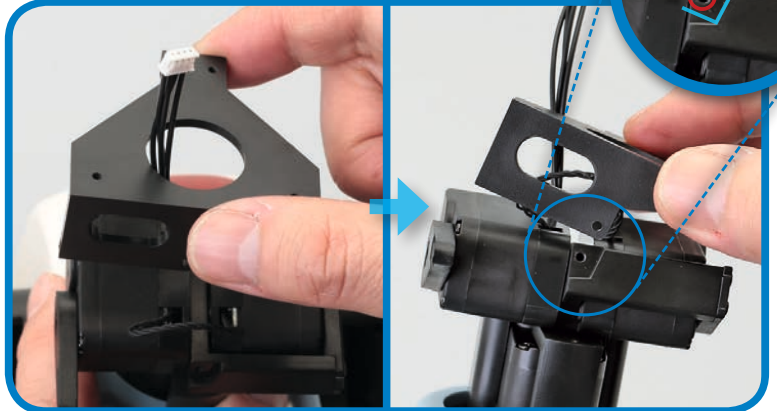


- 6** Once you are happy with the fit, use two M2 x 6mm pan-head screws to secure the parts.

ADDING THE HEAD BASE



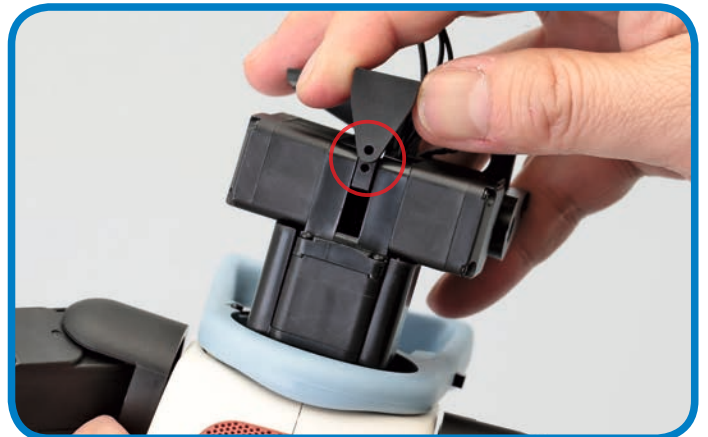
- 7** Fetch the head base and line it up to the top of Robi's head servos and frame, oriented as shown.



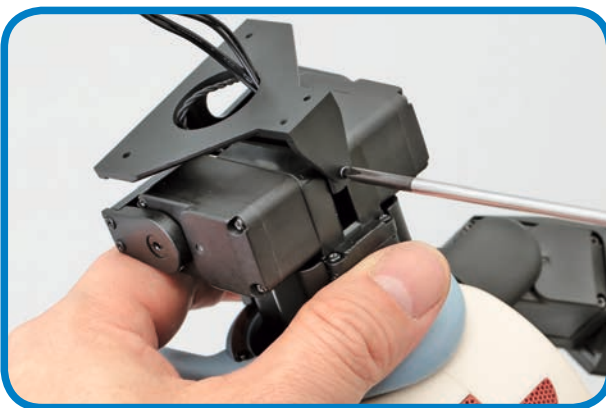
- 8** Lower the base, running the VR cable through the wide hole in its centre. The hole and corner in the centre of the base's backward-facing side, indicated by the red circle and blue lines (inset), should fit neatly with the corresponding ones on the head lever.



- 9** Insert and lightly tighten an M2 x 6.2mm pan-head screw into the hole. Make sure the base is able to swivel slightly on the screw's unthreaded section – do not overtighten the screw. (The shorter screws are very similar, so be sure to use the right one.)



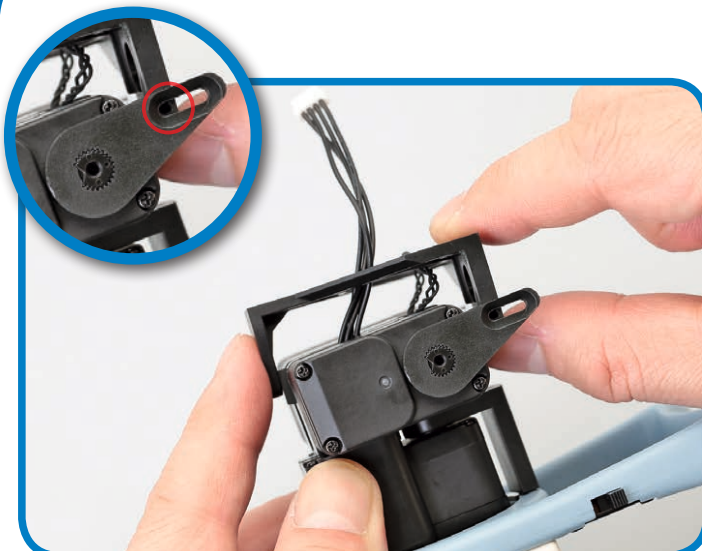
- 10** Now align the hole in the triangular front tab of the head base with the circled hole beneath it.



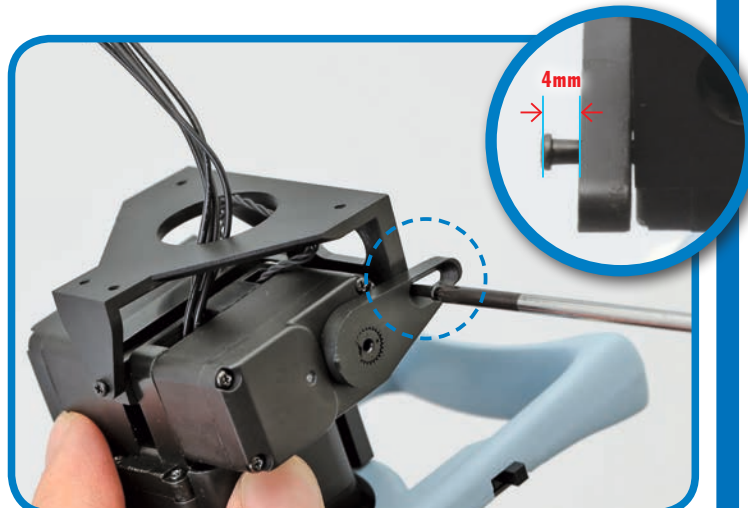
- 11** Secure the base with an M2 x 6.7mm pan-head screw, again taking care not to overtighten it. Check that the base can rotate, and that you have selected the right screw.



- 12** The head base is designed to tilt from side to side, as shown above. If it cannot, or if the movement is stiff, the two screws inserted in Steps 9 and 11 are fastened too tight, so loosen them a little and check the movement again.



- 13** Level the head base so that the circled hole (inset) on its rear left edge lines up with the elongated slot in the left-hand servo horn.



- 14** Fit an M2 x 12mm pan-head screw through the slot and into the hole marked in the previous step. Turn this into the head base, but stop when there is 4mm of the unthreaded section left between the screw head and the servo horn (inset).



- 15** The 4mm length of screw left is used to lift the side of the head base up or bring it down, to tilt the base from side to side. Gently test this a few times by hand to make sure that it moves freely, and if not, go back and check the screws fitted in Steps 9, 11 and 14, loosening them a little if necessary.

Assembled body and head servos



Remember, to see all of Robi's assembly stages, including this one, visit the ModelSpace web page at:

www.model-space.com

Re-testing the head servos



Now that the head servos have their respective horns fitted and are linked up to the head base, it will be easier to see them in action, so you might like to run the test (described in the previous stage) again.



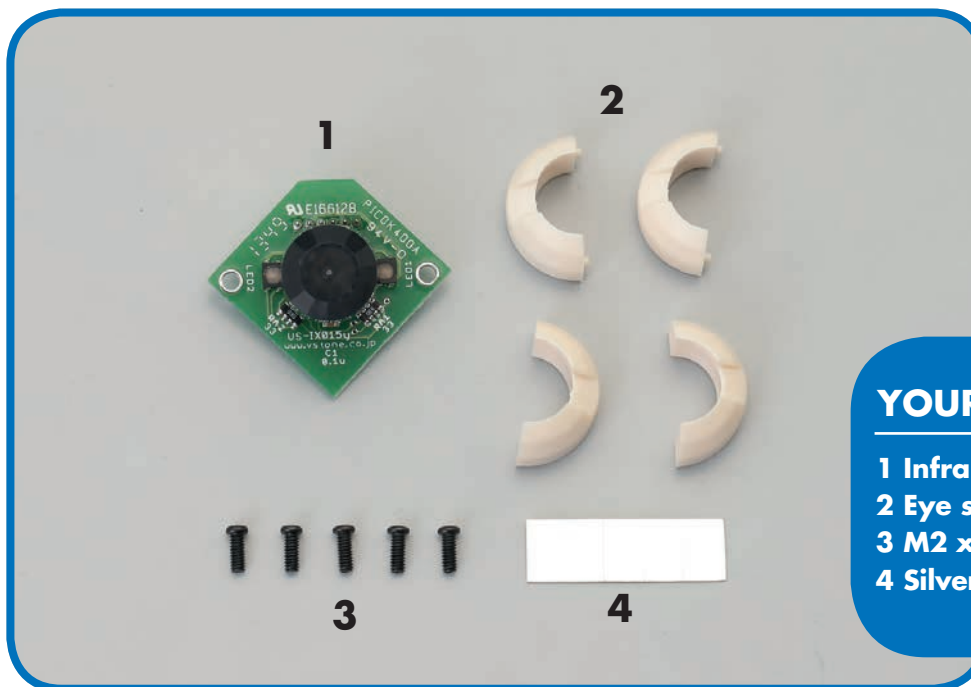
STAGE 62: FIT ROBI'S FIRST INFRARED EYE SENSOR

In this session, you will dismantle Robi's head, then fit the first of the infrared sensors into the socket of one of his eyes.

This stage sees you fit the first of the electronic components that will be concealed inside Robi's head. The infrared eye sensor will connect to Robi's CPU, relaying sensory information of his surroundings to it and enabling him to 'see' what is in front of him.

Before doing this, you will need to dismantle the previously built head to gain access to its internal parts. When you do this, be sure to make a careful note of which screw goes where, because many of them look very similar.

PARTS TO BE ASSEMBLED

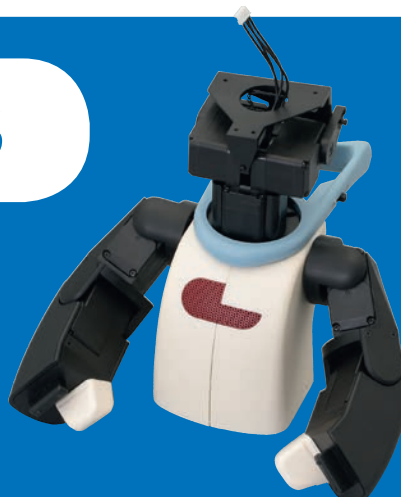


YOUR PARTS

- 1 Infrared eye sensor
- 2 Eye sensor frames x 4
- 3 M2 x 5mm screws x 5 (one spare)
- 4 Silver stickers x 5 (one spare)

SAVED PARTS

You will be using parts from the previous stages, so be sure to have these to hand before beginning.



Body, neck and head assembly (Stage 61)



Robi's head (Stage 60)

FITTING THE SERVO HORNS



ATTENTION!

The M2 x 4.5mm screw used to hold the ear panel in place is very slightly smaller than those used for the ear base (M2 x 6mm). The difference is so small it may be difficult to see with the naked eye, so it is a good idea to keep these separately once removed, and make a note of which is which, to make your job easier when you refit the parts.



1 Unscrew the single screw holding Robi's left ear panel in place. Remove the panel, and **keep the screw**.



2 Now remove the three screws holding the left ear base to the rest of the head. **Again, keep the screws.**



3 Now remove the single screw holding Robi's right ear panel in place. Again, keep the screw carefully.



4 Now repeat Step 2 for the right ear base.



5 Go back to the left side of Robi's head, and remove the seven screws holding the left head frame in place. Again, keep the screws, making a note of where they came from.



6 Leave the two screws securing the left eye in place.



7 Turn the head and remove the two circled screws.



8 You will now be able to remove Robi's face.



9 Remove the three circled screws.



10 Now lift away the top of Robi's head.



11 Remove the two circled screws, this time at the rear of Robi's head.



12 Lift the back of Robi's head away from the assembly.



13 You can now go back and remove the two screws holding the left eye in place.



14 Carefully remove the eye.



15 Turn the head and remove the screws holding the right eye in place.



16 Remove the right eye.



17 Now look into the head frame assembly from above, and remove the four screws holding the left and right frames to the neck mounting.



18 You have now fully dismantled Robi's head, ready to be worked on in the coming steps. The only part you now no longer need is the neck mounting, which you can discard.

FITTING THE SILVER STICKERS



- 19** Take one of the eyes (they are both the same, so it doesn't matter which), and gently push its black iris out from its socket.



- 20** Look inside the eye socket and locate the two screw holes, outlined in red above. The silver stickers supplied in this stage will go over these areas, to ensure that a solid ring of silver lines the eye socket.



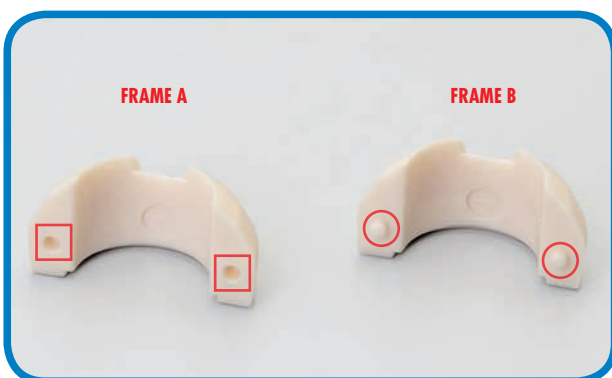
- 21** Fit a silver pad over each of these areas. Using tweezers will help if you find it tricky to do with your fingertips.



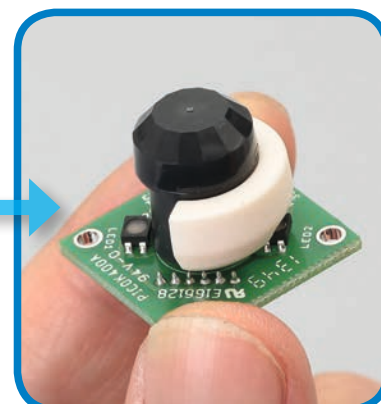
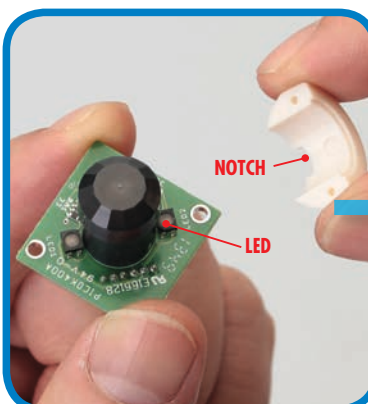
- 22** Your assembly should look like this, the sticker creating a solid ring of silver that lines the inside of the eye socket.

THE EYE SENSOR FRAMES

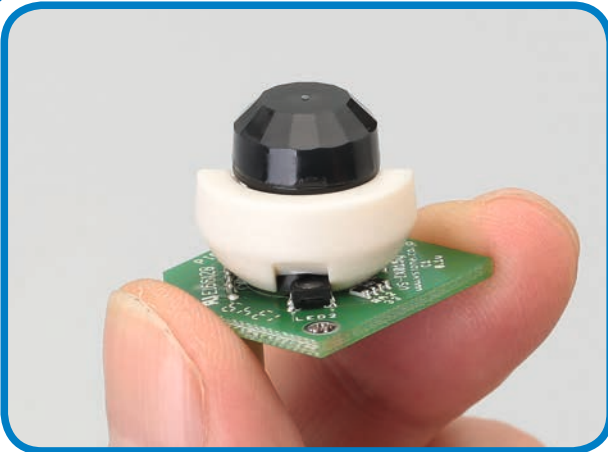
Only hold the exposed circuit board of the eye sensor by its sides!



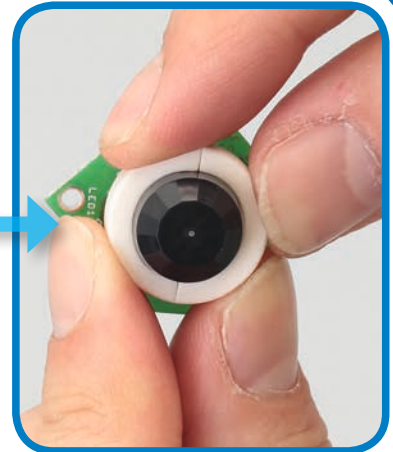
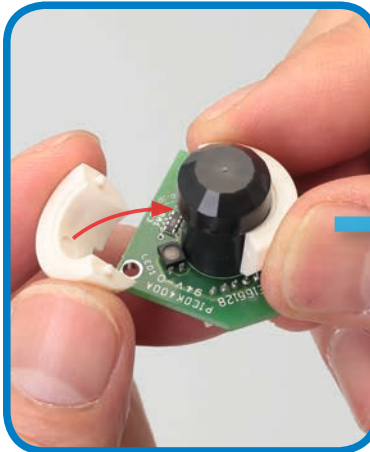
- 23** Familiarise yourself with the eye sensor frames. Each comes in two parts, A and B. Frames A have holes on their ends, which will hold the locating pins of frames B, as seen above.



- 24** Take a frame A and fit it onto the base of the infrared eye sensor supplied in this stage. The notch will accommodate one of the sensor's LEDs, so align the part exactly as shown.



25 Your assembly should now look like this.

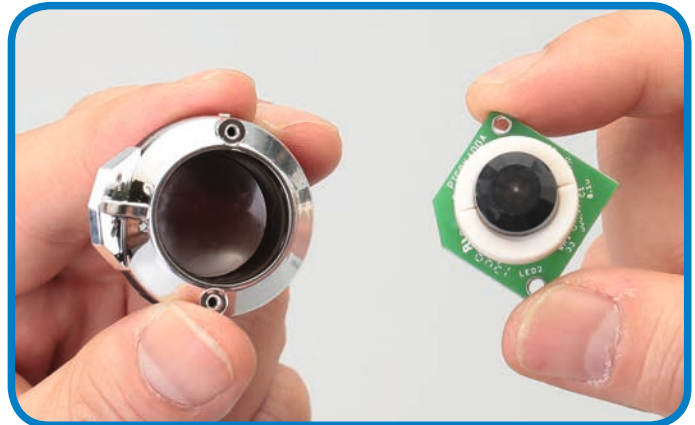


26 Add a frame B to the assembly as shown, with its locating pins fitting into the corresponding holes of frame A. Press the parts firmly together.

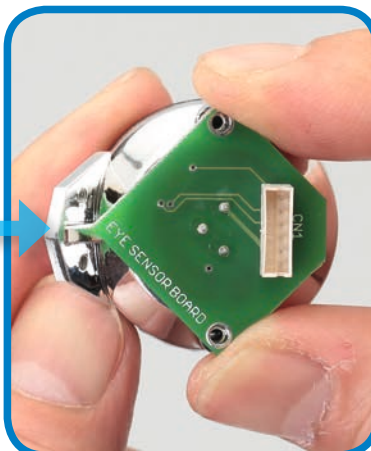
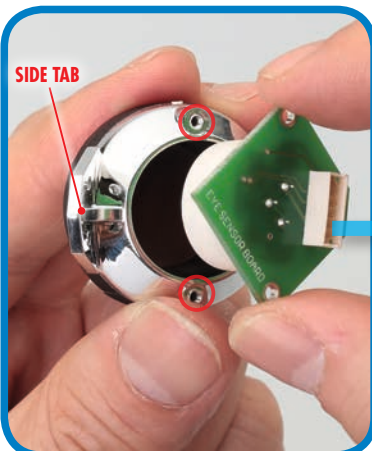
FIT THE EYE SENSOR



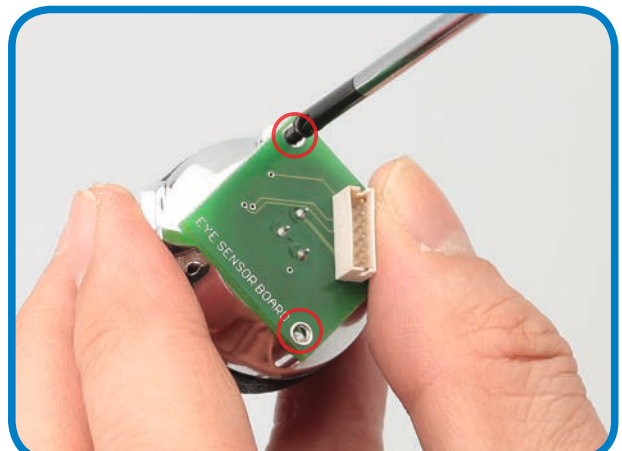
27 Replace the iris removed from its socket in Step 21.



28 Line up the back of the socket with the eye sensor, orienting the parts exactly as shown above.



29 Carefully fit the sensor into the socket. The two screw holes in the sensor's backing should align with those of the socket, with its uncut corner next to the socket's side tab.



30 Use two M2 x 5mm screws to secure the sensor to the eye socket.

FITTING THE HEAD FRAMES



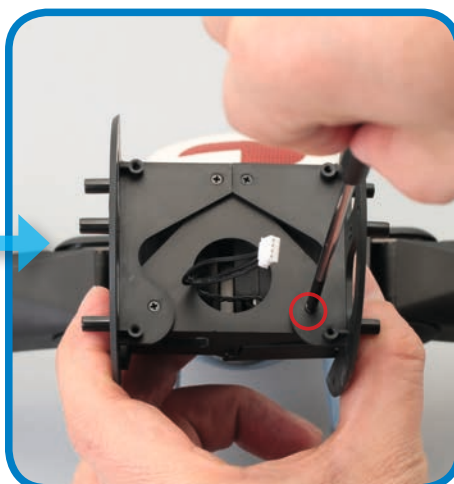
- 31** Fetch the head frames removed in Step 17 and line them up to the head base, facing away from you as shown. The left frame has the shorter arms.



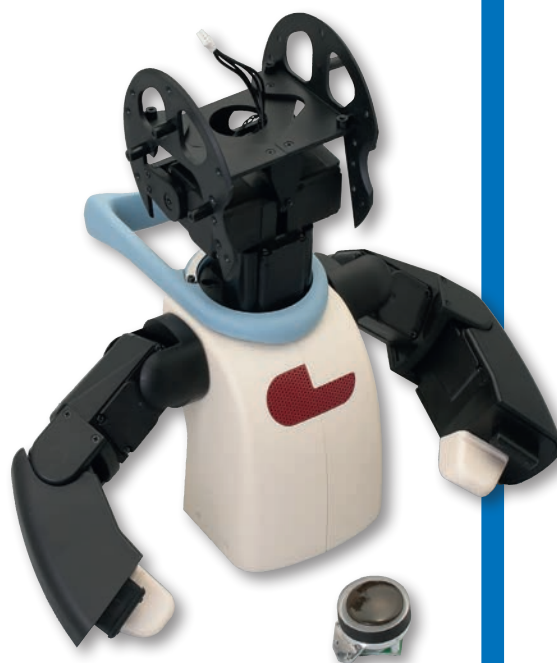
- 32** Use two of the M2 x 4.5mm countersunk screws removed in Step 15 to attach the left frame to the head base.



- 33** Use the other two screws removed in Step 15 to secure the right head frame.



Assembled head and eye



Storing the parts

Store away the eye sensor frames, screws and remaining silver stickers supplied in this stage until they are needed, and keep the delicate eye sensor assembly in a safe place where it is protected from knocks, scratches and static electricity.



Remember, to see all of Robi's assembly stages, including this one, visit the ModelSpace web page at:

www.model-space.com

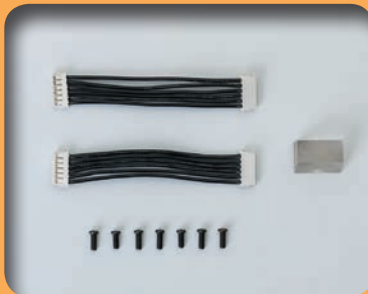
COMING IN PACK 17

Your next **FOUR** complete Stages, as Robi starts to come to life!



Stage 63
Fit Robi's second eye sensor

PARTS PROVIDED
Robi's second infrared eye sensor.



Stage 64
Wire Robi's eye sensors

PARTS PROVIDED
Eye sensor cables and mouth light housing.



Stage 65
Connect up Robi's eyes and mouth

PARTS PROVIDED
The LED board that will control Robi's eye and mouth lights.



Stage 66
Continue wiring Robi's head

PARTS PROVIDED
Mouth LED and infrared cables.